

Main Memory Management in Real-World Operating Systems



CSE 4501 : Operating Systems

Assigned to :

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Comparative Analysis of Main Memory Management: **Windows vs. Linux**

Abstract :

This assignment compares how **Windows 11** and **Ubuntu 22.04 (Linux)** manage main memory. Both OS are widely used but differ significantly in design philosophy, internal architecture, and approach to memory handling. Windows 11 is a proprietary OS developed by Microsoft which emphasizes user experience and backward compatibility. Ubuntu 22.04 is an LTS release of the open-source Linux distribution, designed for flexibility, stability, and control. The comparison will cover key aspects such as memory allocation and deallocation strategies, protection mechanisms, contiguous vs. non-contiguous memory management, and techniques for handling fragmentation. In doing so, this analysis highlights how the design goals of each OS shape their technical trade-offs between performance, complexity, and reliability. The discussion will also link these real-world strategies with theoretical concepts studied in the Operating Systems course.

1. Design Goals Behind Memory Management

Criteria	Windows	Linux
Goal	Maximize performance for user-facing apps with graphical UI; ensure backward compatibility ¹	Flexibility, modularity, and efficiency in multitasking and server workloads ¹
Focus	GUI, multi-user multitasking, compatibility with legacy apps	CLI-heavy, better control, open-source experimentation ¹
Implication	Emphasizes compatibility & responsiveness ¹	Emphasizes efficiency, customizability & performance ¹

2. Allocation and Deallocation Strategies

Aspect	Windows	Linux
Strategy	Uses demand paging ² , working sets, and prefetching	Uses demand paging with copy-on-write ² and slab/slub allocators
Paging	Uses pagefile.sys for virtual memory	Uses swap space (file or partition)
Deallocation	Handled by memory manager & garbage collector (for apps like .NET)	Managed by kernel with reference counting and page freeing logic

3. Memory Protection Mechanisms

Aspect	Windows	Linux
Mechanism	Uses virtual memory with separate address spaces for each process, DEP (Data Execution Prevention) ³ , ASLR	Virtual memory isolation, ASLR, access permission bits, and SELinux for security policies
User vs Kernel Space	Clear boundary enforced	Same, but more configurable

4. Contiguous vs Non-Contiguous Memory Allocation

Aspect	Windows	Linux
Approach	Non-contiguous with paging & segmentation	Non-contiguous using paging, slabs, and buddy system ⁴
Why?	To reduce fragmentation and support large apps	Efficient use of memory in multitasking environments

5. Fragmentation Handling

Type	Windows	Linux
Internal	Managed using heap managers like Low Fragmentation Heap (LFH)	Slab allocator handles it by grouping similar objects
External	Handled via paging and heap tuning	Buddy system ⁴ and virtual memory layout help reduce it

6. Trade-offs Made

Trade-off Area	Windows	Linux
Performance	Prioritizes Smooth GUI Performance	Prioritizes process control and server optimization
Complexity	More complex due to backward compatibility and GUI considerations	Simpler (in kernel-space), but more tuning options
Reliability	High (built-in crash handling)	High

7. Theoretical Concepts vs Real-World Practices

- The paging and segmentation theories are applied in both OSES to manage memory efficiently.
- Fragmentation handling is done using non-contiguous allocation techniques.
- Memory protection through separation of user and kernel spaces is directly derived from OS theory.

References :

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<https://www.webnotes.com/what-is-dep-and-how-to-disable-it-in-windows-11/>

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